

Quantum Mechanics III

Homework 10

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Topic

Continuous Measurement and Quantum Trajectories

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Continuous Measurement and Quantum Trajectories. State all assumptions and conventions.

$$d\rho = \mathcal{L}(\rho)dt + \sqrt{\eta}\mathcal{H}[c](\rho)dW$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$dY_t = \sqrt{\eta}\langle c + c^\dagger \rangle dt + dW$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$E[\rho_c(t)] = \rho(t)$$

Show enough intermediate work that another student can reproduce the calculation.